Nicolas Hanna

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Education

Technical University of Munich **Munich, Germany**
Master of Science, Computational Science and Engineering Oct 2023 – July 2026

- Interdisciplinary 120 ECTS program combining scientific computing, numerical analysis, mathematical modeling, numerical simulation, optimization, and high-performance simulation development.
- Developed expertise in numerical algorithms, parallel computing, scientific programming, software design, and computational methods for solving large-scale engineering and scientific problems.
- **Master's Thesis:** Optimized a sampling-based motion planner for a ROS2-controlled F1TENTH autonomous race car.

Lebanese American University **Byblos, Lebanon**
Bachelor of Engineering, Mechanical Engineering Sep 2019 – Sep 2023

- Graduated with Distinction with a GPA of 3.56/4.00 (1.6 German).
- **Thesis:** Developed a novel renewable heat-exchanger design using phase-change material for thermal energy storage to reduce operational costs and environmental impact. Designed the system in SOLIDWORKS and performed computational analysis in ANSYS.

Work Experience

Technical University of Munich **Munich, Germany**
Student Research Assistant Oct 2025 – Sep 2026

- Developed a high-fidelity digital twin of a commercial stepped-frequency ground-penetrating radar (GPR) antenna for electromagnetic simulation and experimental validation for non-destructive testing.
- Implemented a full-waveform inversion (FWI) algorithm in Python to calibrate the digital twin by estimating material properties that minimize discrepancies between simulated and measured radar responses.
- Engineered and automated ANSYS mechanical simulations of large bridges in Germany to enable non-destructive testing with the help of sensor placement.
- Automated and optimized redundant and repetitive tasks using Python, decreasing the runtime of used functions by 80%.
- Designed in CAD and 3D-printed custom specimen holders and experimental fixtures to enable reproducible radar measurements and validation of the numerical model.

Technical University of Munich **Munich, Germany**
Master's Thesis: Motion Planning for Autonomous Vehicles Jan 2026 – Jul 2026

- Achieved a 2.6 times speed-up by refactoring the motion planning algorithm of the autonomous RoboRacer vehicle using SIMD vectorization.
- Formulated a novel differentiable spatial sampling method for motion planning based on previous research, generating apex-aware paths that significantly decreased overall lap times.
- Integrated an end-to-end differentiable driving stack combining sample selection with differentiable MPC using Python and ROS2 on an NVIDIA Jetson platform.
- Built modular ROS2 node architectures that allowed seamless swapping between open-source gym and physical hardware environments.
- Deployed and validated the software stack on a physical RoboRacer chassis using an OptiTrack motion-capture system on a circuit for live hardware inference.

INDEVCO Phoenix Energy **Safra, Lebanon**
Renewable Energy Engineering Intern Jun 2023 – Sep 2023

- Completed a multistage internship project simulating real-world off-grid energy demand and evaluating appropriate renewable-energy solutions.
- Used SOLIDWORKS to design and simulate steel support structures for solar panels and modeled photovoltaic-system energy output using renewable-energy software.
- Drafted preliminary single-line diagrams based on client specifications for power-flow studies and system optimization. The designs were subsequently reviewed and finalized by an electrical engineer.
- Supported a multidisciplinary team of four engineers during on-site solar-system maintenance and preliminary site assessments for technical consultations.

Projects

Predictive Goalkeeper Rotating Arm — Designed and implemented a control system using MATLAB/Simulink and LabVIEW to tune a PID controller for a DC motor-driven rotating arm. The arm was tasked with intercepting a moving ball equipped with an accelerometer. Deployed the system on a real-time microprocessor for hardware integration and testing.

Smart Door Lock — Designed and built a solenoid-based door-locking mechanism controlled by a PIN code and a concealed laser-sensor trigger. Implemented the system logic in MATLAB and LabVIEW and deployed it on a real-time NI myRIO microprocessor. Used I2C communication for the laser sensor and digital I/O for the keypad. Powered all components through a battery system to ensure portability and uninterrupted operation during power outages.

Turbulent Flow with HDF5 Output — Worked in a team of four to implement efficient large-scale data storage using the HDF5 format for turbulent-flow simulations written in C++. The solution reduced runtime by 30% and file size by 50%. Parallelization was implemented using OpenMPI.

Project Manager — Participated in a two-week summer academy organized by leading German universities. Contributed to the project-management team, overseeing the development of a physics-based video game using the open-source Godot engine and C++. Helped coordinate a team of approximately 20 participants and manage around 50 major project tasks.

Volunteer Experience

Jannat Bladi Foundation — **Beirut, Lebanon**
Humanitarian Volunteer — Aug 2020 – Sep 2020

- Helped clean and restore streets in Beirut following the explosion of 4 August 2020.
- Assisted in establishing a temporary kitchen at a nearby hospital to provide meals for volunteers and residents affected by the explosion.

Skills

Programming Languages	C, C++, Python, MATLAB, Java, G-code for CNC
Robotics and Control	ROS2, Simulink, LabVIEW, PID control, NI myRIO
Engineering Software	SOLIDWORKS, Inventor, Fusion360, ANSYS, AutoCAD, HAP, gprMax, Paraview
High-Performance Computing	OpenMP, MPI, HDF5
Development Tools	Git, Linux
Office Software	Microsoft Word, Excel, PowerPoint
Languages	English (Native), Arabic (Native), French (B2), German (A1.1)